TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES – MANILA

Smart Trash Bin Sensor:
A Prototype for Cleaner Communities

Science, Technology, and Society

GEC 007 / CPE21S2

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I. INTRODUCTION

Overflowing trash bins are a common problem in communities like Barangay Santa Ana, Pateros, creating sanitation and environmental health risks when waste spills into public spaces. To address these operational and public health concerns, this study presents a Smart Trash Bin Sensor Prototype equipped with ultrasonic fill-level detection and a touchless automated lid. The development of this prototype aligns directly with United Nations Sustainable Developments Goals 3 (Good Health and Well-being) by minimizing touch contact with contaminated surfaces and providing timely warnings or alerts to lessen health risks associated with overflowing waste. Furthermore, the prototype also supports SDG 11 (Sustainable Cities and Communities) by enhancing localized neighborhood sanitation, preventing litter accumulation, and reinforcing public waste management infrastructure (*Goal 11: Sustainable Cities and Communities - the Global Goals*, 2024).

II. OBJECTIVE OF THE STUDY

The main objective of this study is to develop a sensor-based trash bin prototype to improve localized waste management and sanitation. Specifically, the study aims to:

1. Determine the fill level of a trash bin using an ultrasonic sensor to detect when the bin is nearly full.
2. Develop an alert system using an Arduino Uno and a buzzer to warn users when the trash reaches a predetermined fill level.
3. Develop an automatic lid mechanism using a second ultrasonic sensor and a servo motor to open and close the lid when a hand or object is detected, provided that the bin is not full.
4. Evaluate the functionality of the prototype in detecting the trash level, activating the alarm, and controlling the automatic lid mechanism;
5. Demonstrate the potential contribution of the prototype to preventing overflowing trash bins and promoting cleanliness and public health; and
6. Assess the social acceptance and practical feasibility of integrating the Smart Trash Bin within Barangay Santa, Ana, Pateros.

III. SIGNIFICANCE OF THE STUDY

The project Smart Trash Bin Sensor demonstrates how technology can support timely garbage collection. Specifically, it aims to help the following:

1. **Households**. The hands-free lid reduces contact with the bin, which helps keep hands clean.
2. **Students**. It promotes awareness of proper waste disposal.
3. **Schools**. It promotes awareness of proper waste disposal and the importance of keeping the community clean and healthy.
4. **Barangays**. It demonstrates how technology can support timely garbage collection.
5. **Future Researchers.** It serves as a reference for improving sensor-based waste management devices and developing additional features.

IV. SCOPE & LIMITATIONS

This study focuses on the development and evaluation of a Smart Trash Bin Sensor Prototype designed to address overflowing waste bins in Barangay Santa Ana, Pateros. The prototype integrates an ultrasonic sensor to monitor waste fill levels, a touchless automated lid to promote hygienic waste disposal, and localized audible alarms to alert users when the bin reaches its capacity. The study aims to support community sanitation, reduce direct contact with potentially contaminated trash bin surfaces, and minimize waste spillage in public areas. The specific scope and limitations of the project are outlined as follows:

Hardware and Technological Constraints. The prototype is limited to the use of an ultrasonic sensor for waste-level detection, a touchless lid mechanism, and a localized buzzer for audible alerts. It does not include Wi-Fi connectivity or a SIM800 module for sending remote notifications or SMS alerts to barangay personnel. Its operation is also subject to power supply availability and the capacity of the light-duty servo motor used to control the lid.

Scope of Prototype Development and Testing. The study covers the design, development, and testing of the prototype's functionality, particularly its waste fill-level detection, touchless lid automation, and audible alert system. The evaluation focuses on the performance of these features under testing conditions and does not extend to the full implementation of a community-wide smart waste management system.

Role as a Supplementary Waste Management Tool. The system serves as a supplementary tool to support the existing waste collection and management services in Barangay Santa Ana, Pateros. It is not intended to replace existing municipal waste management infrastructure, collection schedules, or personnel.

Maintenance and Behavioral Factors. The effectiveness of the prototype depends on proper maintenance, including timely trash bag replacement to prevent tearing due to excessive waste weight, as well as regular inspection of its electronic components. Proper waste disposal practices and community cooperation are also necessary to prevent waste from accumulating around bins that are already full or whose lids are closed.

V. METHODOLOGY

This study employed a qualitative approach alongside hardware development to ensure that the prototype was designed based on the needs of the community. The process involved preparing the technical specifications, assembling the prototype, and conducting a semi-structured key informant interview with the Barangay Captain of Barangay Santa Ana, Pateros. This study includes only one participant: the Barangay Captain.

In this study, purposive sampling was used due to the participant having a better knowledge of the experiences relating waste management, sanitation, and overflowing trash bins around the specified area. The participant will be interviewed through a semi-structured interview.

In this study through using qualitative analysis to identify and interpret the interview participants’s response regarding this study. The interview gathered information about existing

waste disposal practices, identified areas where waste frequently accumulates, such as Esperanza and San Roque, and assessed the feasibility and acceptability of using an automated trash bin in the community. The response was analyzed to identify the community's needs and concerns, which served as a basis for determining the prototype's practical applicability in the community and its design constraints.

VI. DATA GATHERING & RESULTS

Below are the key findings from the semi-structured interview with the local barangay captain provide critical operational insights for integrating automated sensor technologies into community waste workflows:

Existing Municipal Waste System. The barangay of Sta. Ana maintains an active, structured waste management system where garbage collection begins early in the morning (around 6:00 AM). Local community officers (*Tanods*) deploy as early as 3:00 AM to 4:00 AM to supervise waste placement along main thoroughfares and ensure areas remain clean immediately after collection trucks pass. Supplemental surveillance via closed-circuit television (CCTV) cameras is also utilized to monitor real-time waste disposal behaviors across the community.

Bulky Waste Challenge. While standard municipal collection effectively manages routine household waste, significant challenges arise from oversized, heavy, or non-standard materials such as furniture (e.g., beds/mattresses), timber/wood, and stones. Standard waste collection trucks do not pick up these items during regular daily runs. Instead, special collections are scheduled for specific days like Fridays. However, residents often dispose of these bulky items days in advance (such as on Wednesdays), causing waste to accumulate and remain staged along public sidewalks for extended periods before pickup.

Perception of Automated Notifications and Maintenance. Interviews evaluated community feedback regarding automated online or text alerts for full bins or heavy waste. Local officials noted that while dedicated monitoring personnel already provide active coverage, a key

operational issue is replacing full trash bags in a timely manner. If full bags are not replaced promptly, they risk tearing under the weight of accumulated garbage. Nevertheless, the barangay captain emphasized that active barangay monitoring personnel help ensure residents do not scatter waste around full bins.

Identified Possible Deployment Locations. To maximize the operational value of smart trash bins, community leaders recommended deploying prototypes in identified high-accumulation hotspots in their barangay such as Esperanza, San Roque, and surrounding boundary areas. In these locations, automated fill-level alerts and touchless lids can serve as an effective supplementary tool to support existing local community officer (*Tanod*) monitoring and prevent overflow in public areas.

VII. DEFINITION OF TERMS

This study defines the following technical and operational terms to clarify the meaning and functions in the development of the Smart Trash Bin Sensor Prototype. The following terms are defined below.

Technical Terms

1. **Smart Trash Bin Sensor Prototype** – a technology-based waste container equipped with electronic sensors and automated components.
2. **Arduino Uno** – a microcontroller board used to control electronic components through programmed instructions.
3. **Buzzer** – an electronic component that produces sound signals.
4. **Ultrasonic Sensor** – a sensor that measures distance using high-frequency sound waves.
5. **Servo Motor** – an actuator that rotates to a controlled position based on an input signal.

Operational Terms

1. **Smart Trash Bin Sensor Prototype** – a developed prototype that monitors waste levels and supports automated lid operation and notification functions.

2. **Arduino Uno** – serves as the main controller that processes sensor readings and coordinates the buzzer, and servo motor.
3. **Buzzer** – provides an audible warning when the trash bin reaches the programmed fill-level threshold.
4. **Ultrasonic Sensor** – measures the distance between the sensor and the waste surface to estimate the trash bin's fill level.
5. **Servo Motor** – moves the trash bin lid up to 90 degrees to the programmed open or closed position as directed by the control system.

VIII. PROPOSED PROTOTYPE

The proposed prototype is a flip-top trash bin consisting of two ultrasonic sensors, an Arduino Uno microcontroller, a servo motor, a buzzer, and a power supply. The first sensor (Sensor A) is placed on the front of the bin and detects a hand or object approaching the lid. The second sensor (Sensor B) is mounted on the inside back wall of the trash can, facing forward across the container, and measures distance horizontally or diagonally across the upper portion of the bin. The servo motor is attached at the lid hinge and moves the lid.The Arduino Uno reads both sensors and decides what to do:

1. Sensor B measures distance across the upper portion of the bin. As trash accumulates and rises to the height where Sensor B is mounted, sound waves bounce directly off the trash. If the measured distance reaches or falls below the set threshold (10 cm or less), the bin is considered full, triggering the buzzer alarm and keeping the lid closed.
2. If the bin is not full, the Arduino checks Sensor A. When it detects something within about 20 cm, the servo swings the lid open.
3. If the lid stays open for a few seconds, then the servo closes it again.
4. If once the lid is closed, Sensor B takes a new reading and the cycle repeats.

Component	Quantity	Purpose
Arduino Uno R3 with USB Cable	1	Main Controller
HC-SR04 Ultrasonic Sensor	2	Hand Detection (A) and Fill Level Detection (B)
SG90 micro servo (180°)	1	Opens and closes the lid
Active Buzzer	1	Full-bin Alarm
Breadboard and Jumper Wires	1	Electrical Connections
Flip-top trash bin	1 set	Prototype Body
5V Power Supply (Power Bank)	1	Power Source

Table 1. Components of the Smart Trash Bin Sensor Prototype

The table above lists the necessary hardware components required to construct the smart trash bin prototype.

Component	Pin	Arduino Uno
Sensor A (hand)	VCC/GND	5V/GND
Sensor B (fill level)	TRIG/ECHO (Inside Back Wall)	D2/D3
Sensor A (hand)	VCC/GND	5V/GND
Sensor B (fill level)	TRIG/ECHO (Inside Back Wall)	D4/D5
Servo SG90	Red/Brown	5V/GND
Servo SG90	Orange (Signal)	D9
Buzzer	+/-	D8/GND

Table 2. *Pin Connections of the Components to the Arduino Uno*

Table 2 outlines the specific wiring connections between the sensors/actuators and the Arduino Uno microcontroller for proper system operation.

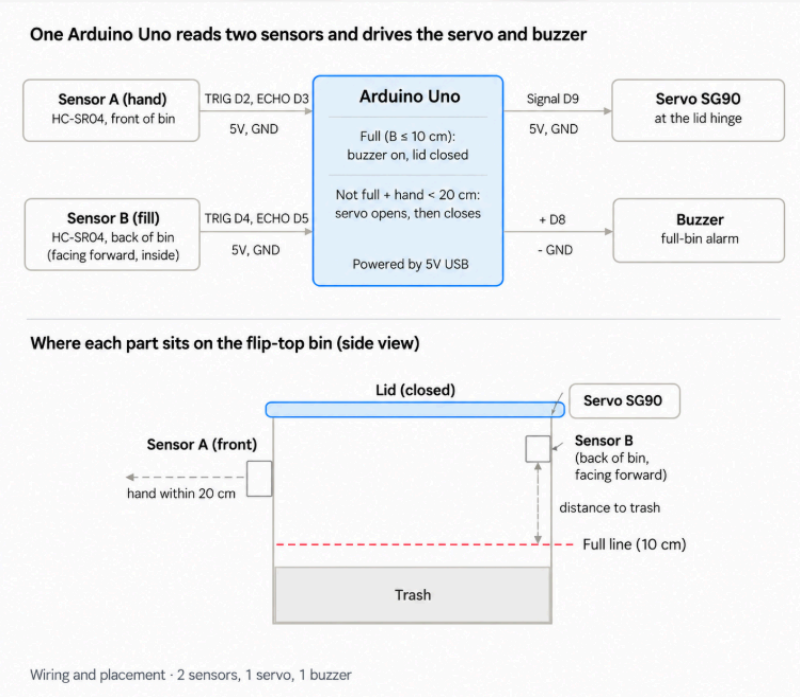


Fig.1. *Block and Wiring Diagram of the Smart Trash Bin Sensor*

In Figure 1, Sensor B, mounted on the inside back wall of the trash can facing forward, checks the fill level first. It measures distance horizontally or diagonally across the upper portion of the bin. As trash accumulates and rises to the height of Sensor B, sound waves bounce directly off the trash to cross the set distance threshold of 10 cm or less, automatically detecting that the bin is full. Upon detection, the Arduino triggers the buzzer alarm and keeps the lid closed.

If the bin is not yet full, Sensor A, at the front of the bin, checks for a hand. When a hand enters the range of Sensor A at 20 cm, the Arduino signals the servo motor to open the lid itself. After a few seconds, the servo motor closes the lid again.

Reference Designs

The photos below are reference designs used as a blueprint for the lid, servo, and front sensor layout. They are not our finished prototype.

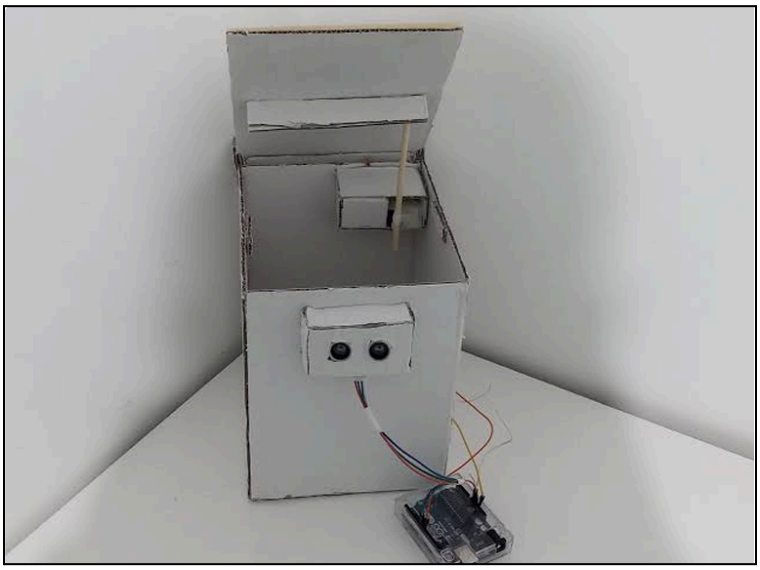


***Fig. 2**. Cardboard Bin with a Servo-Driven Lid and a Front-Mounted Ultrasonic Sensor*



***Fig. 3.** The Front Sensor Detects a Hand or Object so the Lid Can Open*

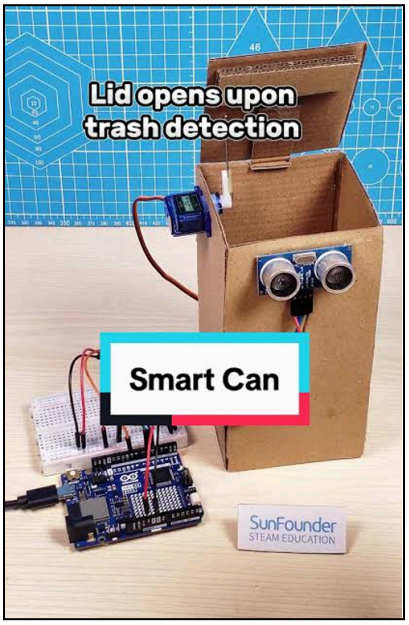


Fig. 4. SunFounder "Smart Can", Where the Lid Opens upon the Trash Detection

IX. CONCLUSION

The Smart Trash Bin Sensor Prototype was developed to explore how electronic components can support waste management and sanitation in Barangay Santa Ana, Pateros. The design integrates ultrasonic sensors for hand and waste-level detection, an Arduino Uno for system control, a servo motor for automated lid operation, and a buzzer for audible alerts.

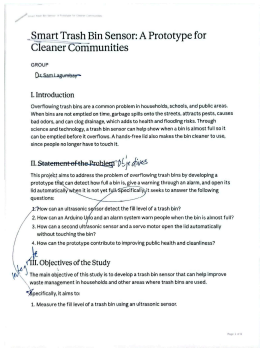
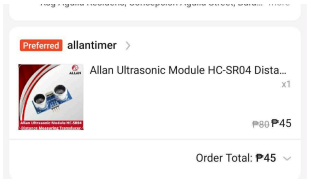
The interview findings revealed that the barangay already has established waste collection and monitoring practices, although bulky waste disposal and the timely replacement of full trash bags remain concerns. The prototype may complement these practices by providing a local fill-level warning and touchless lid operation. Further functional testing is necessary to establish its accuracy, reliability, durability, and practical usefulness in actual community conditions.

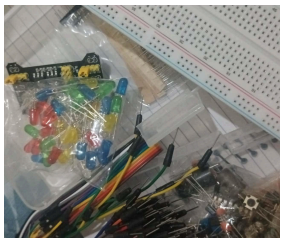
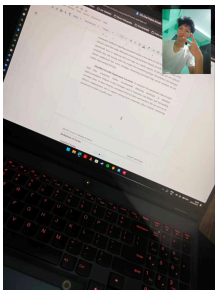
X. RECOMMENDATIONS

This study reveals the potential of the Smart Trash Bin Sensor as a technology-based solution for improving waste management and promoting cleanliness in the community. Thus, the following recommendations are hereby presented:

- Users should ensure that the prototype has a reliable power supply, regularly check its functionality, and empty the bin when the buzzer indicates that it is nearly full.
- Schools and barangays may consider using the prototype to promote cleanliness, proper waste disposal, and awareness of technology-based waste management solutions.
- Waste collectors should maintain regular garbage collection schedules and follow proper waste disposal procedures, as the prototype cannot automatically empty the bin or send remote notifications.
- Future researchers should improve sensor accuracy through proper calibration and placement, explore stronger servo motors and reliable power sources, and consider adding continuous monitoring, wireless notifications, odor control, and waste segregation to enhance the prototype's functionality.
- The prototype should undergo an evaluation for testing the accuracy of the sensor, durability of the prototype, reliability of the automatic lid, and the effectiveness of its alert system.

XI. DOCUMENTATION

Assigned Task	Assigned Member/s	Date	Image	Remarks
First Online Meeting	Gregorio Elento Sardin Basinillo	10/11/26	None	Held an online meeting via Google Meet to brainstorm project ideas.
Drafting Document & Request Letter	Gregorio	10/17/26	None	Prepared the initial project proposal paper and the barangay request letter.
Reviewing Initial Document	Gregorio Elento Sardin Basinillo	10/20/26	 Smart Trash Bin Sensor: A Prototype for Cleaner Communities	Consulted the course instructor to gather feedback on the document draft.
Conceptualizing Prototype Design	Gregorio Elento Sardin Basinillo	10/20/26		Brainstormed and finalized the overall design and layout of the prototype.
Gathering Electrical Components	Gregorio Basinillo	10/20/26		Purchased the servo motor and ultrasonic sensor, and brought jumper wires and a breadboard.

				
Revising Project Paper	Elento	10/21/26		Updated and revised the project paper based on advisor feedback.
Buying Prototype Body Materials	Basinillo	10/21/26		Purchased illustration boards to construct the physical bin structure.
Signing of Request Letter	Gregorio Elento Sardin Basinillo	10/22/26	None	Secured the signature of the course advisor on the barangay request letter.
Building Prototype Body	Sardin Basinillo	10/22/26		Constructed and completed the main physical body of the prototype.
Conducting Barangay Interview	Gregorio Elento	10/24/26		Interviewed key officials from Barangay Santa Ana for community data gathering.
Creating Presentation Slides	Elento	10/27/26	None	Drafted the presentation deck for the upcoming project defense.
Transcribing Interview	Sardin Elento	10/27/26	None	Transcribed the recorded interview and interpreted

Data				key qualitative findings.
Assembling Prototype Hardware & Code	Gregorio	10/24/26 to 10/28/26		Connected all electronic components and finalized the Arduino source code.
Finalizing Project Paper	Elento Basinillo	10/28/26	None	Completed and polished the final report based on gathered interview results.
Finalizing Presentation Slides	Elento	10/28/26	None	Polished and finalized the presentation slides for the final presentation.

XII. APPENDIXES

XIII. A. Request Letter

TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES
MANILA

DATE : October 20, 2026

TO : **Hon. Betty Santos**
Punong Barangay, Barangay Santa Ana
Pateros, Metro Manila

SUBJECT: **REQUEST FOR PERMISSION TO OBTAIN BARANGAY INFORMATION FOR ACADEMIC RESEARCH**

Good day,

We are 2nd Year Computer Engineering students of T.I.P. Manila, currently developing a prototype: A Smart Trash Bin using Arduino Uno

In line with this project, we respectfully request permission to obtain available information from Barangay Santa Ana that is related to waste management in the community, such as garbage collection practices and concerns about overflowing trash bins. The information will only be used as supporting data for our prototype.

Rest assured that all information provided will be used solely for academic purposes related to our research project and will be handled with care. We would be grateful for your approval of this request. For any concern you may contact us through our Mobile No.:+639984721780 or Gmail: mfkgregorio@tip.edu.ph

Thank you for your time and consideration.

Respectfully,

Fausto Kenneth Gregorio
Group Leader

Vale Daniel Sardin
Group Member

Jewel Heart Basinillo
Group Member

Joel Elento
Group Member

Noted by:

Dr. Samuel Lagumbay Jr.
Course Adviser

XIII. B. Interview Questions

Interview Questions

1. **What are the common trash-related problems in your barangay, and how do they affect the community?**
   - *Ano-ano po ang mga karaniwang problema sa basura sa inyong barangay, at paano po naaapektuhan ng mga ito ang mga residente?*

2. **How do you monitor public trash bins and determine when they need to be emptied? Are there existing schedules or protocols for this?**
   - *Paano po ninyo mino-monitor ang mga basurahan sa barangay, at paano po ninyo nalalaman kung puno na ang mga ito? Mayroon po ba kayong sinusunod na iskedyul o protocol para rito?*

3. **What difficulties do you encounter in monitoring trash bins and ensuring timely garbage collection?**
   - *Ano-ano pong problema ang nararanasan ninyo sa pag-monitor ng mga basurahan at pagtiyak na nakokolekta ang basura sa tamang oras?*

4. **Would receiving a message/ notification when a trash bin is full help improve garbage monitoring and collection?**
   - *Sa tingin po ninyo, makatutulong po ba kung makakatanggap kayo ng text o notification kapag puno na ang isang basurahan? Mas mapapadali po ba nito ang pag-monitor at pangongolekta ng basura?*

5. **Where in your barangay would this smart trash bin prototype be most useful?**
   - *Saang lugar po sa inyong barangay sa tingin ninyo magiging pinaka-kapaki-pakinabang ang aming smart trash bin prototype?*

XIII. C. Transcription of Interview

(00:00 - 1:00mins)

Interviewer: So, ano-ano po mga karaniwang problema sa basura sa inyong barangay at pano po ito naapektuhan ang mga residente?
Person 1: Dito sa pinaka barangay kase meron naman nag lilinis dito. 6:00 AM ng umaga lalabas na sila at ang mga basura sa loob hinihila palabas ng bahay at nilalabas na lang nila. Maaga nilalabas ang basura dito, lahat ng basura dito sa Santa Ana at ang pinakaproblema namin ay yung nasa looban.

(1:01 -2:01mins)

Person 1: Pero ang ginagawa naman namin ay nilalabas nila at kinukuha. Dito samin sa lugar ko meron tagabantay. Nagbabantay sila ng alas kwatro ng madaling araw minsan palang 3:00 AM palang andyan na sila, binabantayan nila yung nagkokolekta ng mga basura kaya pagkaalis ng truck ng basura, malinis na ulit ang lahat ganun ang ginagawa dito sa amin.
Interviewer: Next po, pano niyo po mino-monitor mga basurahan sa barangay po? katulad po ng sabi nyo pano niyo po nalalaman kung puno na nga ba ito?
Person 1: Nakikita naman sa CCTV.
Interviewer: Uh, okay thank you po.
Interviewer: Next question po, ano-ano po ang mga problema naranasan ninyo sa pagkuha ng mga basurahang ito at tiyak ba na nakokolekta ng tama ang basura na ito?

(2:02-3:29mins)

Person 1: Kakaunti na lang ang problema namin dito po, yung pag naglabas sila ng malaking basura at yung problema, malaking basura. Halimbawa maglalabas sila ng kama, hindi naman basta kukunin yun may ano yun, may nakatakda kasing araw na kukunin yun. Yun nga. Halimbawa, dito sa amin sa Friday kukunin yung mga iba na

nagtatapon pa lang tuwing Martes o kaya Miyerkules. Kaya, naiimbak ang basura. Halimbawa nag-aano sila ng mga kahoy, nilalabas nila yun hindi kinukuha yun tsaka yung bato hindi kinukuha ng mga magbabasura yun. Kahit hintayin nila pagdating nung dump waste na truck hindi kinukuha ang mga yun. Yun ang problema namin dito, hindi masyado maproblema dito ang basura sa ngayon kasi yung ating mga bantay(*tanod*) na yun, siguro almost twenty ata yun sila e, mga madaling araw nadyan na sila at malayo naman yung nararating nung ating mga bantay(*Tanod)*.

(3:30-4:23mins)

Interviewer: Next question po, sa ginagawa po naming project, sa tingin po ninyo makakatulong po ba ang makatanggap ng text o notification kapag puno na ang mga basurahan o mga basura po na sobrang bigat?
Person 1: Hindi naman, paonline yang sa inyo?
Interviewer: Ang basurahan po ba ay magse-send ng notification po kayo na kung puno na o may mabigat na object na kasama sa basurahan na sobrang laki na?
Person 1: Inaaano pa rin, kami ay inaano sa kanila……kung kami ay di mapapalitan… problem din ’yan kung hindi mapalitan kasi halimbawa minsan punong-puno na yung basura, basura hindi mapalitan mabutas pa, kaya lang hindi nila na ikakalat kasi meron na kami kanya-kanya, kaya ilagay n’yo lang ’yan.

(4:24-5:29:mins)

Interviewer: Saan pong lugar sa inyong barangay ay sa tingin ninyo ay magiging kapaki-pakinabang ang trash bin po namin na magno-notify po sa inyo kung puno na po siya o may mabigat po na gamit?
Person 1: Maglalagay kayo ng trash bin?
Interviewer: I mean, if ever na dito po namin ’yon … gaano po kalaki yung maitutulong na manonotify kayo?
Person 1: Siguro sa “*Esperanza”* diba, Glenn?

Person 2: Po?

Person 1: Trash Bin daw nila saan pwede ilagay? Sa *"esperanza"* diba?

Person 2: Pwede naman po.

Person 1: *"Esperanza"*. Sa bagay don sa "*Esperanza",* dalawa tayong nag aano eh… *"San Roque"* at saka *"Santa Ana"*.

Person 2: Ilan ba ang ilalagay mo?

Person 1: Ilan ba ang ilalagay mo?

Interviewer: Ano lang po sya research prototype pa lang po sya.

Person 1: Ah okay okay.

Person 1: Pag nandiyan na yan marami tayong pag lalagyan. Kung saan natin ipwe-pwesto, doon natin ipwe-pwesto sa laging maraming basura.

Interviewer: Yun lang po, thank you po.

Person 1: Sige sige salamat.

Interviewer: Thank you po.

XIII. REFERENCES

Arduino Making. (2022, May 27). *How to make an automatic trash bin using cardboard*. Arduino Project Hub. https://projecthub.arduino.cc/arduinomaking/how-to-make-an-automatic-trash-bin-using-cardboard-7c4a8d

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